

What the detector actually sees

Pixels on target through the pipeline, and a correction to the figure in the proposal

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1. What the model actually receives

You asked what the detector is being handed, so this starts there and works backwards to the optics. Short version: the numbers in `sizing-calculations.md` §8 were computed for a different sensor than the one we are buying, and the difference matters more to you than to anyone else on the team.

We are buying an **Arducam IMX477** with a 6 mm S-mount lens. Sony specifies it as type 1/2.3: 4056×3040 on a **$1.55 \mu\text{m}$** pitch. The sizing chapter assumes “1/1.8 in. . . **$1.82 \mu\text{m}$** pitch”. Same pixel *count*, different pixel *size* — so resolution on target is not what the chapter says.

Here is the target through the pipeline, at the 40 m search altitude, taking a supine adult as 1.7 m by 0.5 m:

	GSD (cm/px)	Target (px)	Area (px ²)	Of a 640 tile	COCO class
Sensor, native	1.03	165×48	7960	26 %	medium
After $2 \times$ downsample	2.07	82×24	1990	13 %	medium
Chapter figure, 60 m, $2 \times$	3.64	47×14	642	7 %	small

The last row is the one to look at. The 47 px figure currently in the proposal puts the target at 642 px^2 , which is *below* COCO’s small-object threshold of 1024 px^2 . On the sensor we are actually buying, at 40 m, it sits at 1990 px^2 — comfortably in the *medium* band.

That is not a cosmetic difference. Any published recall figure you benchmark against reports AP by that size class, and small-object AP is routinely a large fraction below medium on the same model. If we quote the 47 px number we are implicitly promising small-object performance we do not have to accept.

2. Where those numbers come from

Everything follows from three quantities: pixel count, pixel pitch, focal length. Active area is count times pitch,

$$w = 4056 \times 1.55 \mu\text{m} = 6.287 \text{ mm}, \quad h = 4.712 \text{ mm}, \quad d = 7.86 \text{ mm},$$

which is the type 1/2.3 format Sony quotes. Half-fields are arctangents of half the sensor over f , exactly, with no small-angle approximation:

$$\theta = 2 \arctan\left(\frac{s}{2f}\right) \implies \text{HFOV} = 55.3^\circ, \quad \text{VFOV} = 42.9^\circ, \quad \text{DFOV} = 66.4^\circ.$$

Sensor and ground are similar triangles about the lens, so ground sample distance is just

$$\text{GSD} = \frac{pH}{f} = \frac{1.55 \mu\text{m} \times 40 \text{ m}}{6 \text{ mm}} = 1.03 \text{ cm/px}.$$

One pixel subtends 14.80 mdeg, which is what sets the blur tolerance in §4.

AGL (m)	GSD (cm/px)	Target (px)	Footprint (m)	Frames/target
30	0.78	219	31.4×23.6	6
40	1.03	165	41.9×31.4	8
60	1.55	110	62.9×47.1	12

3. Tiling, and one property worth checking

Tile count depends on pixel count, which did not change, so the inference budget is untouched: at $2\times$ downsample and 2 Hz we still get 12 tiles per frame and 24 inferences per second.

Downsample	Frame	Tiles	Inferences/s	GSD (cm/px)	Target (px)
$1\times$	4056×3040	48	240	1.03	165
$2\times$	2028×1520	12	24	2.07	82

The property worth checking is the overlap. At 640 px tiles and 20 % overlap the shared margin is 128 px, and the target at 40 m is 82 px along its long axis. **The overlap is wider than the target**, so no survivor can be cut by a tile boundary without appearing whole in the neighbouring tile. That holds at every altitude in the table above, and it is the reason the overlap is 20 % rather than something smaller.

4. Blur, which turns out not to bind

Forward motion smears the image by $v t_{\text{exp}}$ on the ground, so holding smear under one pixel means $t_{\text{exp}} \leq \text{GSD}/v$. At 8 m/s:

AGL (m)	Longest exposure (ms)	Shutter
30	0.97	1/1032
40	1.29	1/774
60	1.94	1/516

SYS-45 already mandates 1/1000 s or faster, which clears every altitude. In Indian daylight that pushes ISO down rather than exposure up, so it costs nothing in noise either. Blur is not what will limit recall; target size is.

5. What it costs the flight side

For completeness, since the same substitution moves the coverage numbers in the opposite direction. A smaller pixel behind the same lens narrows the field and finens the GSD in the same proportion, so at 40 m:

At 40 m AGL	IMX477	Sizing chapter	Change, %
Horizontal field of view, $^{\circ}$	55.30	63.20	-12.5
Ground sample distance, cm/px	1.03	1.21	-14.8
Survivor along the long axis, px	164.52	140.11	+17.4
Swath width, m	41.91	49.21	-14.8
Line spacing, m	29.34	34.45	-14.8
Transects per drone	7.00	6.00	+16.7
Sweep time per drone, s	195.75	166.93	+17.3

We gain 17 % more pixels on target and pay 17 % more sweep time — seven transects per drone instead of six. Against 250 marks for detection and an 1800 s budget I think that favours the sensor we are buying, but it was never a decision anyone made.

6. What I need from you

Two things, and neither is urgent this week.

First, if you have benchmarked anything on HERIDAL or SARD, tell me which size band the reported AP came from. If the published number was measured on medium-class targets then our 40 m figure is comparable and the 47 px one was never going to be.

Second, the survey altitude is unresolved — the design point says 60 m, every simulation we have flown says 40 m, and the sizing chapter proposes 40 m pending a recall measurement. That measurement is yours. If you can say what recall looks like at 82 px against 47 px, the altitude decision follows from it rather than from argument, and I can re-baseline the documents once instead of twice.

The prose here is mine; the arithmetic is not. Every figure in this note is computed by `camera_optics.py` and typeset by `camera_optics_tex.py`, both under `tools/sizing-model/`, so the document cannot drift from the model the way the sizing chapter drifted from the parts list — which is the whole subject of the note.